

Free Group

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Problem 1. Let $i : X \rightarrow X$ be an "involution", i.e. a map from the set X to itself with $i \circ i = id_X$. Show that i is a bijection (in the most efficient way you can).

Proof. Let $i : X \rightarrow X$ be an involution thus $i \circ i = id_X$. Thus i has a left inverse, i is surjective (Equivalence Relations Homework). Furthermore since i has a right inverse it is injective (Equivalence Relations Homework). Since i is surjective and injective it is a bijection. $\square$

Problem 2. Now let S be a set with a fixed point free involution $i : S \rightarrow S$ (ie $i(s)$ is different from s for all s in S). By a "word" in S we mean a finite string of elements. We also allow the empty string. This set of words is a "monoid", we can multiply them just by concatenation. It is easy to see that this operation is associative. But it is not yet a group, we have to introduce some cancelation. If w is a word in S , and it contains two consecutive elements $s i(s)$ (note this is also $i(t)t$ for $t := i(s)$), then the "simple reduction" is obtained by removing s and $i(s)$. A word obtained by a series of simple reductions is called a reduction. A word is called reduced, if there is no pair $s i(s)$ of consecutive elements in the word.

Solution. There is no question here $\blacksquare$

Problem 3. Show that every word has a reduced reduction.

Proof. Base Case: All words of length 1 are of reduced form.

Assume that all words with length k has a reduced reduction.

Assume w is a word with length $k + 1$. If w is reduced then w is the word's reduced reduction. If w is not reduced, there is some $xi(x) \in w$ for $x \in S$ then replace $xi(x)$ in w with 1, thus length of w decreases by one and thus length of w is k and thus w has a reduced reduction by our inductive hypothesis $\square$

Problem 4. Show that the reduced reduction is unique.

Proof. Let w_0 be the reduced form of w . It is obtained by a series of cancellations on w . The first case is a pair $x, i(x)$ is cancelled at some step of the sequence. If so we may cancel $x, i(x)$ first. Thus for any cancellation we make we can move it to be step and still get the same word. On the other hand since w_0 is a reduced form $x, i(x) \notin w_0$. Thus at least one of the symbols must be cancelled at a time. If the pair itself is not cancelled, the first cancellation involving the pair must look like $\dots i(x)\mathbf{x}i(\mathbf{x}) \rightarrow \dots i(x)\dots$ or $\dots \mathbf{i}(\mathbf{x})\mathbf{x} \dots \rightarrow \dots x \dots$. This is the same as cancelling the pair $xi(x)$ giving us the first case. Thus the sequence of cancellations commutes thus any sequence of steps with the same elements as the steps to get w_0 for w would still produce w_0 . $\square$

Problem 5. Show this is an equivalence relation, and show each equivalence class contains a unique reduced word.

Proof. 1. **Reflexive:** $\forall a \in W$ let the reduced form of a be a_0 . Then a has the same reduced form as itself which is a_0 .

2. **Symmetric:** $\forall a, b \in W$ if a, b have the same reduced form w_0 . Then b also have the same reduced form w_0 .

3. **Transitive:** $\forall a, b, c \in W$, if a and b have the same reduced form w_0 and b and c have the same reduced form w_1 . Since the reduced form of b must be unique then $w_0 = w_1$ and thus the reduced form of c is w_0 and thus a and c have the same reduced form.

Now we say that two words are equivalent if they have the same reduced reduction. Thus $\forall a, b \in W$, $a \sim b \iff$ they have the same reduced form where $\sim$ is a equivalence relation by above.

Furthermore, $\forall a \in W$, let $[a]$ be the equivalence class of a . Since a must have a unique reduced form by problems 3 and 4, let a_0 be the reduced form of a . Since a_0 is the reduced form of a and a_0 is the reduced form of itself, $a \sim a_0$ and $a_0 \in [a]$. Thus every equivalence class has a reduced form. $\square$

Problem 6. Now take $F(S)$ the set of equivalence classes of words. We give it a multiplication $F(S) \times F(S) \rightarrow F(S)$ by $[a][b] := [\text{red}(\text{red}(a)\text{red}(b))]$ where $\text{red}(a) \text{red}(b)$ means concatenate the reduced version of a and b . Show this is well defined, and makes $F(S)$ into a group.

Proof. $\forall x \in S$, define to be $x_0 := \text{red}(x)$. Let W be the words formed by S .

Well Defined: $\forall a, x \in W$, $\forall a, b \in [a]$ and $\forall x, y \in [x]$, since $a_0 \in [a]$ is the unique reduction of a and b and $x_0 \in [x]$ is the unique reduction of x and y . $[a] * [x] = [\text{red}(\text{red}(a)\text{red}(x))] = [\text{red}(a_0x_0)]$ and $[b] * [y] = [\text{red}(\text{red}(b)\text{red}(y))] = [\text{red}(a_0x_0)]$ and since $[b] = [a]$ and $[x] = [y]$ and $[a] * [x] = [\text{red}(a_0x_0)] = [b] * [y]$. Thus $*$ is well defined.

By definition of the multiplication on $F(S)$, $\forall a, b \in W$, $[a][b] = [\text{red}(\text{red}(a)\text{red}(b))] \in F(S)$ thus $F(S)$ is closed. $[1] \in F(S)$ where 1 is the empty word which is by definition reduced, thus $\forall a \in S$ $[1] * [a] = [\text{red}(\text{red}(1)\text{red}(a))] = [\text{red}(1\text{red}(a))] = [\text{red}(a)] = [a]$ by definition of empty word and equivalence classes similarly $[a] * [1] = 1$. $\forall a \in W$, a_0^{-1} is the word where a_0 is reversed and each "letter" is replaced with its involution. Assume the contrary, a_0^{-1} is not reduced. Then there exists some $i(x)x$ or $xi(x)$ in a_0^{-1} and thus reversing the inverting process there would be $xi(x)$ in a_0 meaning a_0 is not reduced thus forming a contradiction. Thus $\forall a \in W$ $[a] * [a_0^{-1}] = [\text{red}(\text{red}(a)\text{red}(a_0^{-1}))] = [\text{red}(a_0a_0^{-1})] = [1]$ and similarly $[a_0^{-1}] * [a] = [1]$. Thus each element has an inverse.

$F(S)$ is isomorphic to W_r reduced words which is a subset of W and thus since W is a monoid, W_r must have associativity and thus $F(S)$ must also have associativity.

Thus $F(S)$ is a group $\square$

The above is not exactly the standard treatment. Instead we start with a set X , take another set X' with a bijection $b : X \rightarrow X'$. Now we take S to be the disjoint union of X and X' with its natural involution (its b on X and the inverse of b on X'). Now apply the above. We will call this $F(S)$

Now let G be a group and $s : S \rightarrow G$ a map (of sets, S is just a set). We say that $s : S \rightarrow G$ is a free group on S if s is universal for maps to groups, ie if given any other map of sets $t : S \rightarrow G'$, with G' a group, there exists a unique homomorphism of groups $g : G \rightarrow G'$ with $t = g \circ s$ ($\circ$ means composition).

Problem 7. Show that any two free groups on S are canonically isomorphic (meaning there is a distinguished iso, unique in a way you should make precise)

Proof. Let S be a set with free groups G and G' . Thus there exists maps $s : S \rightarrow G$ and $t : S \rightarrow G'$ such that there exists unique homomorphisms $g : G \rightarrow G'$ and $g' : G' \rightarrow G$. Thus since g and g' are unique, then $g' \circ g$ must also be a unique homomorphism from G to G . $id_G \in \text{Hom}_{gr}(G, G)$ thus $g' \circ g$ is a unique homomorphism then $g' \circ g = id_G$. By symmetric logic, $g \circ g' = id_{G'}$ thus g and g' have left and right inverses thus they must be bijections. Thus g and g' are bijections and homomorphism and thus isomorphisms. Thus any two free groups on S are canonically isomorphic. $\square$

Problem 8. Suppose G is free on S , show that $s : S \rightarrow G$ is injective (using just the universal property)

Proof. Let G be a free group on S . Assume the contrary, $s : S \rightarrow G$ is not injective. If $|S| = 1$, then s is forced to be injective and thus a contradiction forms here. However if $|S| \geq 2$, then if s is not injective then $\exists a, b \in S$ s.t. $s(a) = s(b)$ where $a \neq b$. Order the elements of S , $s_1, s_2, \dots, s_n$. Let $t : S \rightarrow \mathbb{Z}/|S|\mathbb{Z}$ such that $s_i \rightarrow [i]$. Since s is not injective, $\exists i \neq j$ for $i, j \in [1, \dots, n]$ such that $s(s_i) = s(s_j)$ and by prehomework this implies $t(s_i) = t(s_j)$ since t factors through s . However $t(s_i) = [i] \neq [j] = t(s_j)$ since $i \neq j$. This violates the universal property and G cannot be free group. Thus a contradiction forms and s must be injective. $\square$

Problem 9. Suppose G is free on S , show that G is generated by $s(S)$ (again, just using the universal property)

Proof. Suppose G is free on S . Let $s : S \rightarrow G$. Let $G' = \langle s(S) \rangle$. Then let $t : S \rightarrow G'$ such that $t = g \circ s$ and $\forall x \in S$ $x \rightarrow s(x)$. By definition of the universal property, there must be a unique homomorphism from $g : G \rightarrow G'$ which for all $x \in S$ takes $s(x) \rightarrow s(x)$. Since G is a free group, s must be injective by problem 8. Furthermore $\forall a, b \in G$, $g(a) = g(b)$ since $g(a), g(b) \in \langle s(S) \rangle$ $g(a) = g(b)$ sequence of $t(f_i)$ for $f_i \in S$. By definition of g $t(f_i) = s(f_i) = g(s(f_i))$ thus the sequence of $t(f_i)$ for $f_i \in S$ turns into a sequence of $g(s(f_i))$ thus a and b equal the same sequence of $s(f_i)$. Thus g is injective. Furthermore for all $a \in G'$, it is by definition a sequence of $s(f_i)$ thus by definition of g $g(s(f_i)) = s(f_i)$ and since g is a homomorphism $\exists a_g \in G$ where a_g is the same sequence of $s(f_i)$ and since g is a homomorphism $g(a_g) = g(s(f_1)) \dots g(s(f_n)) = s(f_1) \dots s(f_n) = a$. Thus g is surjective. Thus g must be a bijection. $\square$

Problem 10. Show that $F(S)$ above is free on S .

Proof. Let G be a group. Let S be a set. Let $s : S \rightarrow F(s)$ be a function where $\forall s \in S$ $s \rightarrow s \in F(s)$. Let $t : S \rightarrow G$. Define a function, $f : F(S) \rightarrow G$ such that $s \rightarrow t(s)$ and $s^{-1} \rightarrow t(s)^{-1}$ and such that for $a, b \in S$ $ab \rightarrow t(a)t(b)$. Thus $t = f \circ s$. For $W, V \in F(S)$, let $red(W) = a_1^{\pm 1} \dots a_n^{\pm 1}$ and $red(V) = b_1^{\pm 1} \dots b_m^{\pm 1}$ and $a_i \in S$ and $b_j \in S$ for all $i \in [1, \dots, n]$ and all $j \in [1, \dots, m]$. $V * W = red(red(V)red(W))$ as seen above. Thus $f(V * W) = f(a_1^{\pm 1} \dots a_n^{\pm 1} b_1^{\pm 1} \dots b_m^{\pm 1}) = t(a_1^{\pm 1}) \dots t(a_n^{\pm 1}) t(b_1^{\pm 1}) \dots t(b_m^{\pm 1}) = f(a_1^{\pm 1} \dots a_n^{\pm 1}) * f(b_1^{\pm 1} \dots b_m^{\pm 1}) = f(red(V)) * f(red(W))$ and since $V \sim red(V)$ and $W \sim red(W)$ $f(red(V)) * f(red(W)) = f(V) * f(W)$ thus f is a homomorphism. Thus $F(S)$ is free on S by the universal property. $\square$